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Data from the *All of Us* research program reinforces existence of activity inequality



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Large-scale and detailed analyses of activity in the United States (US) remain limited. In this work, we leveraged the comprehensive wearable, demographic, and survey data from the *All of Us* Research Program, the largest and most diverse population health study in the US to date, to apply and extend the previous global findings on activity inequality within the context of the US. We found that daily steps differed by sex at birth, age, body characteristics, geography, and built environment. Quantifying activity inequality using the modified Gini index, we found a strong correlation with obesity prevalence ($R^2 = 0.804$) and a moderate correlation with perceived walkability ($R^2 = 0.426$) and the activity gender gap ($R^2 = 0.385$). This study demonstrates the value of digital health technologies in exploring and understanding public health practices while highlighting the need to examine complexities, including biopsychosocial factors that may contribute to activity inequality.

While the health benefits of engaging in physical activity are well-recognized¹, several studies have unveiled inequality in physical activity based on gender or income^{2–4}. Previously, insights into a population's physical activity levels were gathered primarily through self-reported questionnaires. However, technological advancements now enable physical activity monitoring using mobile and wearable devices in free-living conditions. These streams of data gathered continually on a daily or even minute-by-minute basis can be shared and analyzed in aggregate to gain insights into the overall physical activity levels of the population.

A recent large-scale digital health study on 717,527 smartphone users across 111 countries demonstrated a correspondence between activity inequality and obesity and a city's walkability, with findings of gender differences in activity and obesity⁵. Such findings, if found to be robust and replicable within the population of the United States (US), can serve as a quantifiable benchmark for assessing disparities in physical activity in the US and their relation to sex, age, geographic location, built environment, and other potential key influencing factors. Further, such measures can be used as metrics of success in evaluating the effects of policy changes on public health.

Activity inequality studies have yet to be replicated within the US due to a lack of available digital health data from independent and disease-agnostic studies. Large research initiatives across the US, such as the Behavioral Risk

Factor Surveillance System (BRFSS)^{6,7} and the National Health and Nutrition Examination Survey (NHANES)⁸, have collected behavioral and activity information using self-reported phone surveys and research-grade activity monitors. Yet, activity monitoring was only collected during a 7-day monitoring period and is thus likely to suffer from high variance and noise. To enable detailed exploration of US activity levels at the population scale, large-scale digital monitoring data is critical.

The *All of Us* (AoU) Research Program, an initiative of the National Institutes of Health (NIH) launched in 2018, strives to build a robust ecosystem of communities and researchers to deliver the largest and most diverse publicly-available biomedical dataset with a focus on populations usually under-represented in biomedical research in the US⁹. Specifically, the program accumulates multiple streams of health-related information such as electronic health records (EHRs), genomics, physical exam measures, participant surveys, and digital monitoring data (e.g., Fitbit fitness trackers) from ≥1 million Americans.

In this work, we sought to evaluate the utility of AoU data for replicating two seminal digital health studies^{5,10} that demonstrated activity inequality and its relationships to geographic and walkability measures, which we derived from AoU's Social Determinants of Health Survey¹¹. We also assessed the relationship between activity inequality and key health outcomes (e.g., obesity).

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Results

Overall participant characteristics

Of the 15,620 AoU participants with Fitbit data available at the time of our analysis, we included 7403 participants with over 4.1 million days of monitoring prior to the COVID-19 pandemic (January 2016 to December 2019) from 47 states, the District of Columbia, and other US Territories (Flow diagram in Supplementary Fig. 1). The participants included in our analytical cohort had a median age of 53 [IQR (37, 64)], a median BMI of 28.2 kg/m² [IQR (24.5, 33.1)], and a daily steps average of 8330 (Table 1). Nearly 68% and 81% of the participants included in the analysis were Female and White, respectively.

Variability in daily steps across US states

Based on average daily step counts across different US states between 2016 and 2019, we observed that AoU participants in cooler regions (i.e., average temperature $\leq 60^\circ\text{F}$) engaged in higher levels of physical activity compared to those in warmer regions (average temperature $>60^\circ\text{F}$) (Fig. 1a). The five most active states based on average daily steps were New York, Colorado, Illinois, Massachusetts, and California (Fig. 1a). The five least active states were Louisiana, Tennessee, Arizona, New Hampshire, and Texas. We also examined activity intensity levels using definitions of ≥ 150 min of moderate-intensity aerobic physical activity (MVPA) or ≥ 75 min of vigorous-intensity aerobic physical activity (VPA) per week, respectively (see Methods)¹. In states with the warmer average temperatures, 28% of people did not meet the recommended levels of MVPA or VPA, whereas this percentage was around 10% less in regions with cooler temperatures (Supplementary Fig. 3a).

To better characterize the differences in activity between relatively “active” and “inactive” US states using daily steps, we compared the distribution of daily steps in the two most active states (New York and Illinois) and the two least active states (Texas and Arizona) (of states with $n > 100$ participants). All states exhibited unimodal distributions (Fig. 1b), and the modes (i.e., the peaks) were lower than the respective means (Fig. 1b). Such positive skewness of the distributions suggests that the range of daily steps is wider among the active individuals in each state. In addition, we found that the mode of daily steps for Arizona (7196) was higher than in Texas (6839), whereas the mean of daily steps in Arizona (7568) was lower than in Texas (7648). The higher mode in Arizona compared to Texas indicates that the distribution skewness varies across states and, hence, reinforces the utility of activity inequality evaluation, which can uncover activity variance within a region⁵.

We next compared the overall distribution of physical activity across New York, Illinois, Texas, and Arizona while normalizing the distribution by each state’s mode (i.e., the most common activity level), enabling a comparison of how the activity is distributed within each state (Fig. 1c). After normalization, we observed that 79% of the population falls within 50% of the mode (the region between the light gray vertical dashed lines for all curves) in Arizona and 72% in Texas. This difference in the distribution curves can be translated into a metric of activity inequality, which is measured by the Gini coefficient of the daily steps (see Methods) and characterizes the systematic difference level in physical activity. Quantifying our visual inspection from Fig. 1c, activity inequality in Texas (0.223) is higher than in Arizona (0.217) because Texas has a slightly more asymmetric distribution. Interestingly, while New York was one of the most active states and Arizona was one of the least active, the two states display a similar distribution (Fig. 1c), indicating a similar variance of activity around the most commonly achieved daily steps of each state.

Variability in the gender-activity gap across US states

We hypothesized that the skewness in each state’s activity distribution (Fig. 1b, c) may be attributable to gender differences in activity levels given the findings by Althoff et al.⁵. Across all populations, males had approximately 1200 more daily steps on average than females. The difference in daily steps between males and females was statistically significant in all states with more than 20 males and females each (see Methods; student one-sided *t*-test;

p-value < 0.05), except for Washington (Supplementary Fig. 5). We also observed that these activity disparities increase as activity quantiles increase, with the largest difference in daily steps between the most active males and females (Fig. 2c). Comparing Texas and Arizona, the two least active states from Fig. 1b, the disparity in typical activity levels (defined by the mode) between males and females was approximately 2.3 times greater in Texas (2885 steps) than in Arizona (1241 steps) (Fig. 2a). In Texas, the distribution of daily steps in males is right-skewed while the activity distribution of females is left-skewed when compared to the state’s baseline activity (Fig. 1b). Overall, not only do typical daily steps differ across states (Fig. 1a) but so do the distributions of daily steps, which is attributable in part to the underlying difference in activity levels between females and males. In fact, 38.5% of activity inequality in the US was explained by the activity differences between males and females (Fig. 2b). As we found between Texas and Arizona, states with larger gaps in activity between males and females have higher overall activity inequality.

Obesity and physical activity

Obese and overweight individuals took 1963 and 531 fewer daily steps than those with normal body mass index (BMI), respectively. Consistent with previous findings⁵, we found that the fraction of obese individuals ($\text{BMI} \geq 30$) in each state increases with higher activity inequality ($R^2 = 0.639$ [$N = 18$ states], $R^2 = 0.804$ [excluding outliers: $N = 16$]) (Fig. 3a). With every 0.01 increase in the value of the activity inequality metric, there is a 5% increase in the fraction obese. We further hypothesized, based on prior global findings⁵, that this relationship may be due to females displaying a more rapid increase in obesity with a decrease in daily steps. While we found that obesity prevalence does indeed increase as daily steps decrease for both genders, we saw no difference in the slope of this relationship (Fig. 3b). Therefore, the higher obesity rate in states with higher activity inequality cannot be entirely explained by the lower-activity females as the obesity prevalence increases equally in both males and females as average daily steps decrease.

Walkability, biopsychosocial factors, and physical activity behaviors

Walkability is a commonly used metric for gaining insight into a built environment, which may act as an extrinsic factor influencing physical activity^{1,12}. The perceived walkability of a neighborhood (see “Methods”), which can range from 0 to 18, did not display a strong linear relationship with average daily steps ($R^2 = 0.0132$) (Supplementary Fig. 6), but we observed a moderate association between perceived walkability and activity inequality ($R^2 = 0.4263$, *p*-value < 0.01) (Fig. 4a). When average steps, by the time of day, were examined across neighborhoods with highest (14 to 18) and lowest (0 to 4) perceived walkability scores, we observed high activity from 8 AM to 6 PM during weekdays (typical US business hours) and from 9 AM to 5 PM on weekends, with a gradual decrease in activity throughout the day for both weekdays and weekends (Fig. 4b). These trends are consistent with previous findings⁵. We also found that the two peaks in steps that typically occur on weekdays during commute hours exhibit distinctly higher step counts in neighborhoods with high walkability (294 vs. 259 and 315 vs. 242; Fig. 4b). The distribution of steps taken early in the morning was similar between neighborhoods with high walkability ($n = 5$) and low walkability ($n = 5$), but individuals in the high walkability neighborhoods had higher average steps throughout the evening after 5 PM during weekdays. During weekends, neighborhoods with high walkability continued to have a higher activity level after 10 AM as compared to neighborhoods with lower walkability.

The observation that certain times of day (e.g., during commute hours on weekdays) have higher step counts in more walkable neighborhoods points to the possibility that public transit and pedestrian infrastructures may be drivers of physical activity. Given existing health disparities and the established concept of activity inequality^{5,13,14}, we sought to understand whether the relationships we uncovered between the built environment and physical activity are consistent across demographic subgroups. To enable this comparative analysis, we defined a new metric: the walkability-activity

Table 1 | Demographic characteristics of the 7403 participants

Characteristic	Participants		
	Male	Female	Other or Unknown
Gender, <i>n</i> (%)	2108 (28.5%)	5094 (68.8%)	201 (2.7%)
Age (years), mean (SD)	54.28 (15.93)	49.45 (15.32)	51.45 (15.36)
Age (years), <i>n</i> (%)			
18–34	363 (17.2%)	1165 (22.9%)	38 (18.9%)
35–54	574 (27.2%)	1815 (35.6%)	71 (35.3%)
55–64	516 (24.5%)	1176 (23.1%)	38 (18.9%)
≥65	655 (31.1%)	938 (18.4%)	54 (26.9%)
BMI (kg/m ²), mean (SD)	29.13 (5.75)	29.67 (7.33)	29.73 (6.98)
BMI, <i>n</i> (%)			
<18.5	<20 (<1%)	<20 (<1%)	<20 (<1%)
18.5–24.9	420–530 (20–25%)	1250–1530 (25–30%)	50–60 (25–30%)
25–29.9	860 (40.8%)	1495 (29.3%)	65 (32.3%)
≥30	757 (35.9%)	2089 (41%)	81 (40.3%)
Race, <i>n</i> (%)			
Asian	102 (4.8%)	130 (2.6%)	<20 (<1%)
Black or African American	76 (3.6%)	326 (6.4%)	<20 (<1%)
Middle Eastern or North African	<20 (<1%)	<20 (<1%)	<20 (<1%)
Native Hawaiian or Other Pacific Islander	<20 (<1%)	<20 (<1%)	<20 (<1%)
Not indicated	111 (5.3%)	239 (4.7%)	169 (84.1%)
Other	58 (2.8%)	141 (2.8%)	<20 (<1%)
White	1744 (82.7%)	4243 (83.3%)	29 (14.4%)
Ethnicity, <i>n</i> (%)			
Hispanic or Latino	140 (6.6%)	308 (6%)	<20 (<10%)

Exact participant counts for categories <20 cannot be shown as per the *All of Us* Data and Statistics Dissemination Policy. They are presented either as <20 or in ranges to prevent participant count of less than 20 from being derived from other reported cells or information, including in text, tables, or figures.

gradient (WAG). The WAG characterizes the rate at which average activity levels increase with increasing walkability ($\Delta\text{steps}/\Delta\text{walkability}$).

On average, individuals with normal BMI ($n = 2056$) took 531 steps and 1963 steps more daily compared to individuals in the overweight ($n = 2420$) and obese ($n = 2927$) BMI groups. Among all BMI groups, individuals with normal BMI also consistently recorded the highest average step counts across all levels of perceived walkability, followed by those in the overweight and then the obese categories (based on BMI). Specifically, at a perceived walkability score of 10 (the mean for the study population, indicated as a dashed vertical line in Fig. 4c, d), individuals with normal BMI averaged 9047 steps while those with overweight or obesity averaged 8329 and 6832 steps, respectively. Above a perceived walkability score of 10, the WAG was highest in the normal BMI group (166 steps per unit of perceived walkability), followed by the obese (138) and overweight groups (104), suggesting that the physical activity of the normal BMI group increases most per perceived walkability score out of all BMI groups. Interestingly, below a walkability score of 10, the overweight group initially demonstrates a decrease in physical activity while the obese group displays no change in physical activity with increasing perceived walkability.

A difference in the relationship between perceived walkability and physical activity was also seen across the lifespan. Overall, adults over age 65 ($n = 1647$) took 721 fewer daily steps than adults below age 65. At a perceived walk score of 10, the 35–54 and 55–64 age groups had average daily step counts of 8062 and 8184, respectively. The 65+ age groups and,

surprisingly, the 18–34 age group both displayed significantly lower average step counts at 7732 and 7542 steps. All ages displayed a more rapid increase in step counts after a perceived walkability score of 10. However, the increase in activity with walkability was not as drastic in the older age groups (65 + : 88 steps increase per unit of perceived walkability and Age 55–64: 113) as compared to the younger groups (Age 18–34: 202 and Age 35–54: 237).

Discussion

In this study, we leveraged the unique AoU dataset to assess the replicability of seminal research studies on population-scale physical activity and to uncover new findings in this field. We demonstrated variation in activity levels across geographic regions and demographic subgroups. We also assessed the translatability of a previously proposed activity inequality metric⁵ to enable a better understanding of factors that may affect physical activity and that can be used to support policy decisions such as investing in infrastructure to improve walkability. Our findings provide a foundation for robust evaluation of physical activity across the United States by extracting actionable insights from commercially available wearable device data.

Our replication efforts pointed to challenges in comparing results across different population-level physical activity studies that use different measurement modalities. We found that the AoU's Fitbit-based population mean (2016–2019) was 3560 daily steps greater than the smartphone-measured population average reported by Althoff et al. (2013–2014)⁵. In contrast, we found that the proportion of active individuals (meeting the aerobic guideline of ≥ 150 min/week of MVPA or ≥ 75 min/week of VPA¹) in each state was, on average, 25% higher than in the BRFSS study¹⁰, which employed self-reports. These differences in findings point to the potential role of the timing of the studies, assessment modalities (smartwatch vs. smartphone vs. survey-based self-reports), and/or data processing methods. It has been suggested that smartphone-based step counts tend to be lower than smartwatch-based step counts¹⁵. These, combined with our usage of an HR-based filtering algorithm to minimize adherence bias, may explain why average steps in AoU were higher than those reported by Althoff et al. for US¹⁶. Separately, methods of self-reporting physical activity¹⁰ fail to capture absolute levels of physical activity^{17,18} and may be susceptible to recall and response biases. Also, while the details of Fitbit's algorithm are proprietary, Fitbit notes that both heart rate and step data are used to estimate metabolic equivalents (METs) and that active minutes are earned when activities with ≥ 3 METs are recorded for ≥ 10 continuous minutes¹⁹. Previous validation studies of Fitbit showed comparatively accurate estimates of sedentary behavior but overestimated MVPA under free-living conditions¹⁷. Together, these points may explain why the proportion of active individuals¹ was higher in AoU as compared to BRFSS. The overall physical activity differences across these population-scale studies in the US highlight the need for standardized methods (e.g., measurement modality agreement, harmonized data processing algorithms, consensus definitions, and metrics) to enable both observational and comparative effectiveness studies whose results can be used together to improve meta-analyses in this field.

A relationship between activity inequality and obesity has been previously demonstrated across countries⁵, and we found that the same relationship held true across the AoU cohort. However, while sex differences were previously demonstrated as a mediator of this relationship globally, we did not find this to be true for the AoU data. In the US, we found that neither overall obesity prevalence, which is also consistent with the NHANES study²⁰, nor its responsiveness (i.e., the amount of reduction in obesity rate) to physical activity levels differs between males and females. Our study suggests that other potential underlying factors such as social determinants of health (e.g., healthcare access and quality, education access and quality, social and community context, economic stability), lifestyle, and dietary choices must be additionally explored to explain the relationship between activity inequality and obesity prevalence. In addition, while the AoU Research Program collected BMI through standardized in-clinic physical measurements, Althoff et al. relied on self-reported height and weight to calculate BMI⁵. Self-reported data may lead to an underestimation of BMI compared to in-clinic measurements because individuals tend to

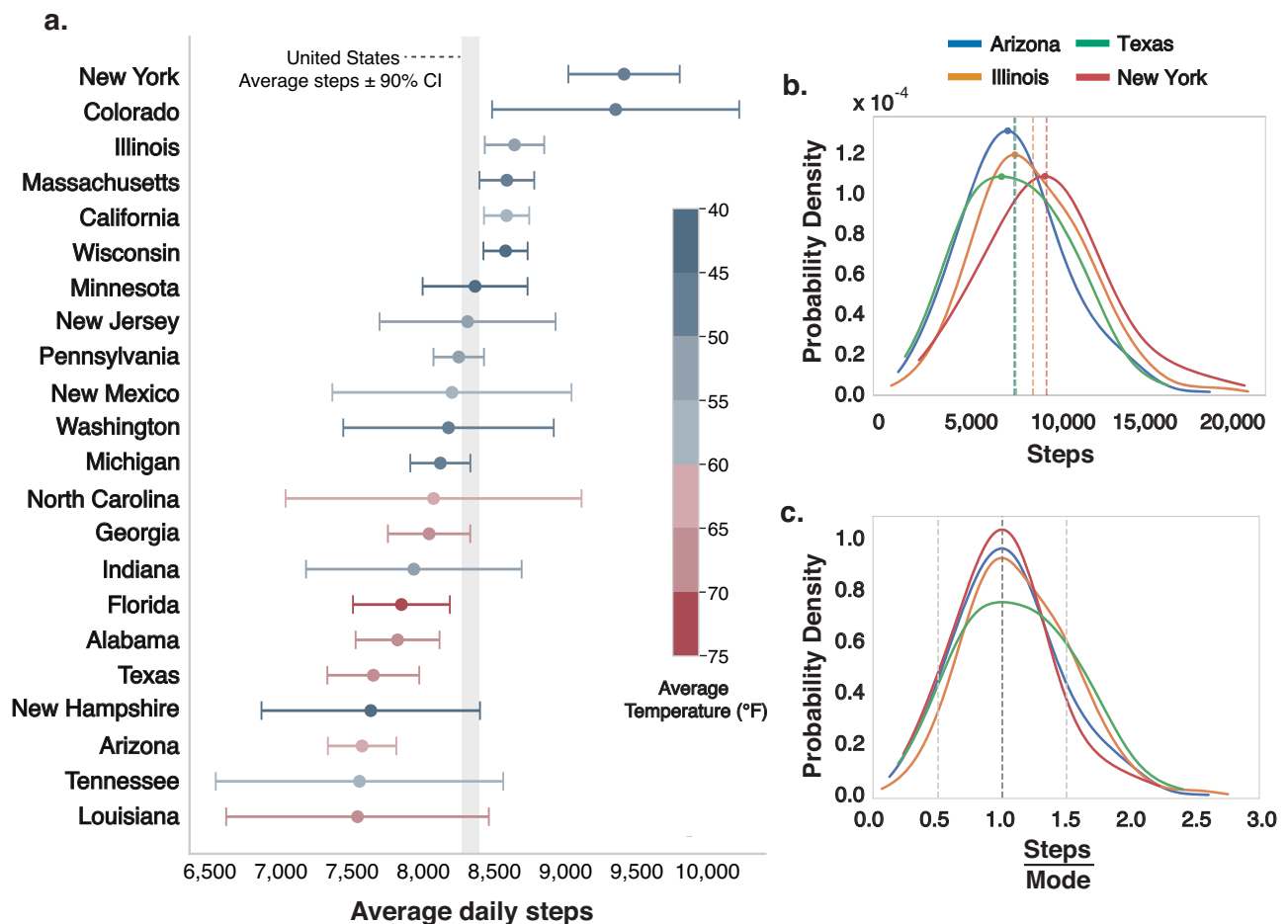


Fig. 1 | Demonstration of activity differences across US states using the AoU cohort. **a** The forest plot illustrates the variation in activity levels across the 22 states included in the analysis. Lower levels of daily steps are typically found in states with higher average temperatures. The gray vertical region indicates the average daily steps of all states (including states with $n \leq 20$). Each error bar indicates the 90% confidence interval. **b** (Figure adapted from Althoff et al. Figure 1⁵) The distribution of two states with high activity (New York and Illinois) and two states with low

activity (Texas and Arizona). New York has the highest mode (9336), and Texas has the lowest (6915). The mode of each state is indicated by dots, and dashed vertical lines correspond to the mean of each state. **c** (Figure adapted from Althoff et al. Figure 1⁵) The distribution plot, normalized by the population mode of steps, reveals that New York and Arizona have similar distributions. The dark gray vertical dashed line indicates the distribution plot normalized to each state's mode, and the light gray vertical dashed lines are placed at 50% below and above the mode.

underestimate their weight and overestimate their height²¹. This methodological difference may account for some of the observed discrepancies between the studies.

Walkability is an important factor considered in urban planning that is closely related to quality of life, health benefits, and sustainability. Previous work demonstrated an inverse relationship between activity inequality and walkability, as measured by WalkScore, which assesses macro-scale features like amenities (e.g., parks and shopping centers) and pedestrian friendliness (e.g., city block length, street intersection density) at the city level^{5,21}. In our study, we were unable to use WalkScore due to the absence of city-level information in the AoU dataset. Instead, we evaluated micro-scale walkability using the Physical Activity Neighborhood Environment Scale (PANES), which captures individuals' perceptions of their local environment. Despite these methodological differences, we found a similar relationship between activity inequality and perceived walkability. This suggests that policies and interventions aimed at both macro-level improvements (e.g., enhancing street connectivity and transit systems) and micro-level changes (e.g., improving neighborhood safety) could help reduce activity inequality.

However, we also found that the rate of increase in physical activity with improvement in walkability varies across BMI and age groups, which may be attributed to various biopsychosocial factors that differ among these groups. This suggests that while universal improvements in walkability can

broadly increase the overall PA level, policy decisions might need to be tailored to maximize benefits across different demographic strata and improve activity inequality. Although macro- and micro-scale measures of walkability are often correlated, they may affect different demographic groups in distinct ways. If city-level information becomes available in the AoU dataset, future studies should investigate the added value of micro-level walkability information, and whether one measure of walkability is more sensitive to the built environment's influence on physical activity within certain demographics. This comparison could provide valuable insights for improving the overall physical activity level and reducing activity inequality.

As a quantifiable metric, activity inequality can systematically measure the uneven distribution of physical activity within groups, gaining an understanding of the extent of these disparities. Further, it provides a baseline for evaluating the effectiveness of targeted interventions and policies over time. For instance, future researchers can explore how activity inequality and its magnitude vary with changing temperatures and geographies, especially during the record heat waves that occurred across the US in the summer of 2024. The quantifiability of activity disparity also allows for targeted and evidence-based interventions. Instead of adopting a one-size-fits-all approach, policymakers can use activity inequality as a metric to allocate resources and interventions that address the specific needs of communities or demographic groups experiencing the highest levels of activity inequality. Overall, activity inequality serves as a data-driven

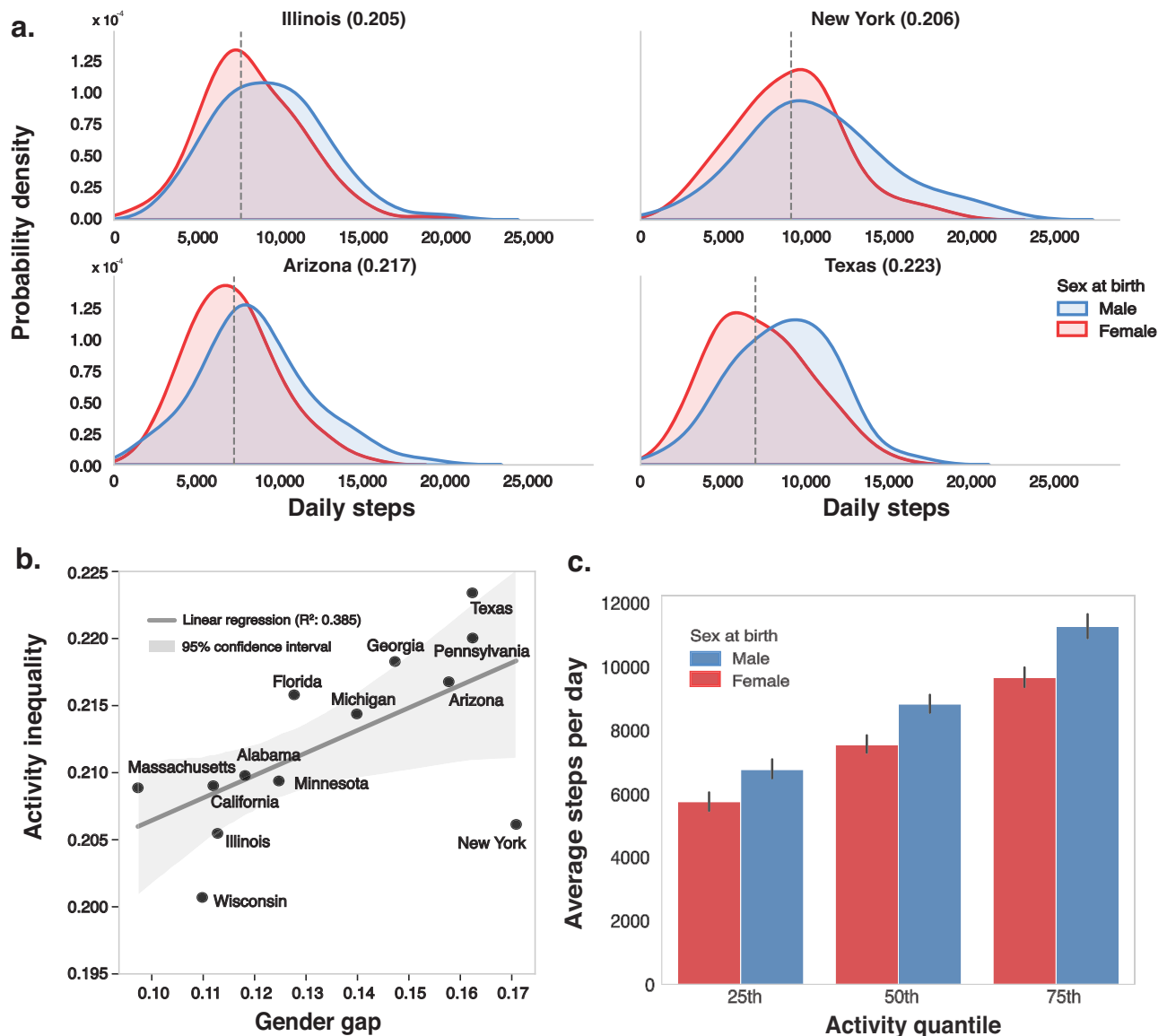


Fig. 2 | Relationship between gender and daily steps, activity inequality, and activity level. (Figure adapted from Althoff et al. Extended Data Figure 7⁵). **a** Probability density curve by sex assigned at birth of two states with high activity (New York and Illinois) and two states with low activity (Texas and Arizona). The vertical dashed lines represent the mode steps of each state. The number in parentheses represents the activity inequality measure of each state. **b** Activity inequality is explained by the gender gap calculated by dividing the difference in

daily steps between males and females by the daily steps of males ($R^2 = 0.385$). The 95% confidence interval is estimated using bootstrapping and is indicated as the light gray bands around the regression line. **c** Association between average steps per day and activity quantile. On average, males take more steps per day than females in all activity quantiles. The 95% confidence interval (i.e., error bar) is constructed using bootstrapping.

foundation for identifying disparities, assessing the impact of interventions, and tailoring strategies to address the unique challenges faced by different populations. By leveraging this metric, public health efforts can become more precise, equitable, and effective in promoting physical activity and improving overall health outcomes.

One limitation of this study is its demographic composition, which was skewed toward individuals who were >50 years of age, overweight, or obese. There was also a significantly higher proportion of female participants in this study cohort (68.8% female, $n = 5094$) compared to the population and to prior studies, including the CDC's BRFSS study conducted in 2017 (55.8%, $n = 223,241$) and in 2019 (54.5%, $n = 203,199$)¹⁰ or the Althoff et al. study (51.1%, $n = 98,971$)⁵. Therefore, it is likely that our estimates, such as the average daily steps in a region or walkability-activity gradient, may be biased toward the highly represented demographics. These discrepancies in the published literature emphasize the importance of accounting for

demographic and contextual factors in analyses and point to the value of additional future work exploring associations between physical activity or activity inequality and external factors.

Further, the variability in sample size (N) across states in the AoU dataset may also limit the generalizability of our findings, particularly in states with smaller sample sizes like Tennessee, New Mexico, or North Carolina. However, the longitudinal nature of the AoU data allows us to better estimate individual baseline physical activity levels compared to the NHANES accelerometry dataset, which is constrained by a shorter, one-week observation period. Aside from the variability in sample size, bring-your-own-device (BYOD) studies can lead to selection bias toward technology users who may differ from the general population²². In an effort to address these concerns, AoU recently launched the WEAR study, which provides participants with a Fitbit Charge 4 or Fitbit Versa 3 device at no cost²³. With the addition of the WEAR study, future researchers can further

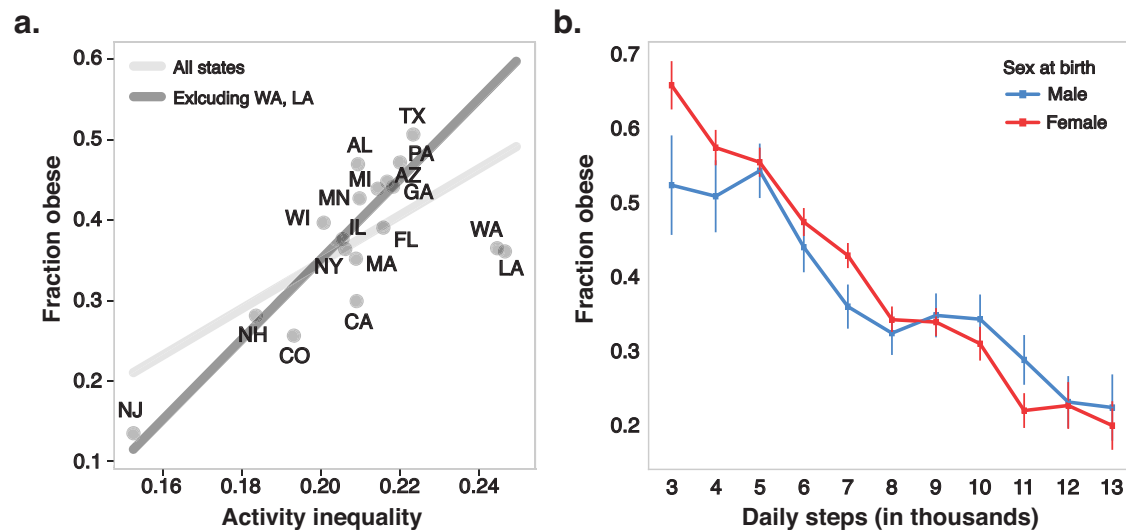


Fig. 3 | Obesity prevalence is associated with an increase in activity inequality. (Figure adapted from Althoff et al. Figure 2⁵). **a** Two linear regression lines were fit: with all states ($R^2 = 0.639$) and without the two outliers that were identified visually — Washington and Louisiana ($R^2 = 0.804$). **b** As daily steps decrease, obesity

prevalence increases. There is no significant difference in this association between males and females, as indicated by the overlapping 95% confidence intervals (i.e., error bars), which is estimated through bootstrapping.

validate the findings of this study with a dataset more representative of the US population for enhanced generalizability.

Additionally, our Fitbit dataset spanned three years (2016–2019), and we assumed that participants did not relocate during this period and had no significant changes in their BMI, given that BMI was only measured once per participant during the AoU data collection period (Supplementary Fig. 7). Also, the AoU's Social Determinants of Health (SDOH) survey we used to calculate the perceived walkability scores was conducted two years after the wearable monitoring period (November 2021–June 2022). We assumed that perception of one's neighborhood did not change significantly between the Fitbit observation period and the time the survey was taken. Future research should account for how potential spatial and temporal changes such as activity changes over time, geographic relocations (between states, cities, or countries), and seasonal and behavioral variations in BMI and activity, may impact the results of this study²⁴.

Our study assessed physical activity mainly with average daily steps, but future studies can leverage other metrics, such as fairly active minutes and/or very active minutes obtained from Fitbit's daily summary report. In addition, while our study focused on obesity as the main health outcome of interest, there are additional physical activity-related health outcomes that remain to be explored. Finally, the Fitbit device model information was not available on the Researcher Workbench at the time of analysis to enable adjustment for possible differences in data generated by different Fitbit device models. We attempted to minimize the impact of this limitation by including participants' activity data only if they had corresponding heart rate data, thus generalizing the findings to Fitbit devices with heart rate compatibility.

Despite these limitations, our study provides insights to advance knowledge surrounding physical activity disparities in the US by leveraging wearable device data. Our study builds upon and departs from prior activity inequality studies in several innovative ways. We applied a heart rate-based filtering algorithm to minimize adherence bias and ensure high-quality physical activity (i.e., daily step counts) data from commercial wearable devices. Our findings on variations in definitions of activity inequality across the US further support the need for equitable interventions to promote physical activity, which can, in turn, reduce the prevalence of obesity and chronic diseases²⁵. We assessed adapting a prior global analysis of activity inequality⁵ to the US using the AoU data and found both consistencies and departures with prior literature. While we found lower activity levels in

females compared to males and the association between activity inequality and obesity to be true, we could not demonstrate the higher reduction in obesity with an increase in steps in females. The differences in findings are most likely due to the demographic differences between datasets and the smaller variance in physical activity and obesity rate among states (i.e., homogenous lifestyle) compared to countries. Future investigations of robust activity inequality measures are needed to better capture activity differences across the US, especially when the data may be imbalanced (e.g., demographics are skewed toward one sex or age group), and to establish a relationship to health outcomes.

Overall, this study explored activity behaviors and activity inequality in the US, paving the way for future research that harnesses activity inequality metrics derived from commercial wearable device data. The activity inequality measure opens future paths for research to explore the intricate connections between activity inequality, health outcomes, and targeted interventions for the US. The research that is enabled by this study could revolutionize how population-level physical activity interventions are conceptualized and implemented in the future.

Methods

AoU Researcher Workbench

The AoU Researcher Workbench provides a secured cloud-based interface²⁶ that allows registered researchers to access and analyze data from various sources, including surveys, physical exam measurements, electronic health records, and wearable devices such as Fitbit. In this study, the Controlled Tier dataset version 7 (C2022Q4R11) was used as it provided granular sociodemographic information (e.g., up to three digits of zip codes) and non-date-shifted Fitbit device data.

In this study, only the authorized authors who completed AoU Responsible Conduct of Research training accessed the data from the Researcher Workbench. All research using data made available via the AoU Researcher Workbench has been designated as non-human subjects by the AoU Institutional Review Board. Since the authors were not directly involved with the participants and Researcher Workbench employs a data passport model, IRB review was not required. Further, all analyses and reporting conducted as part of this study have been performed in accordance with the AoU Data and Statistics Dissemination Policy. The queries, external datasets, and Python code used for statistical analysis are all available in the Featured Workspace as Jupyter Notebooks.

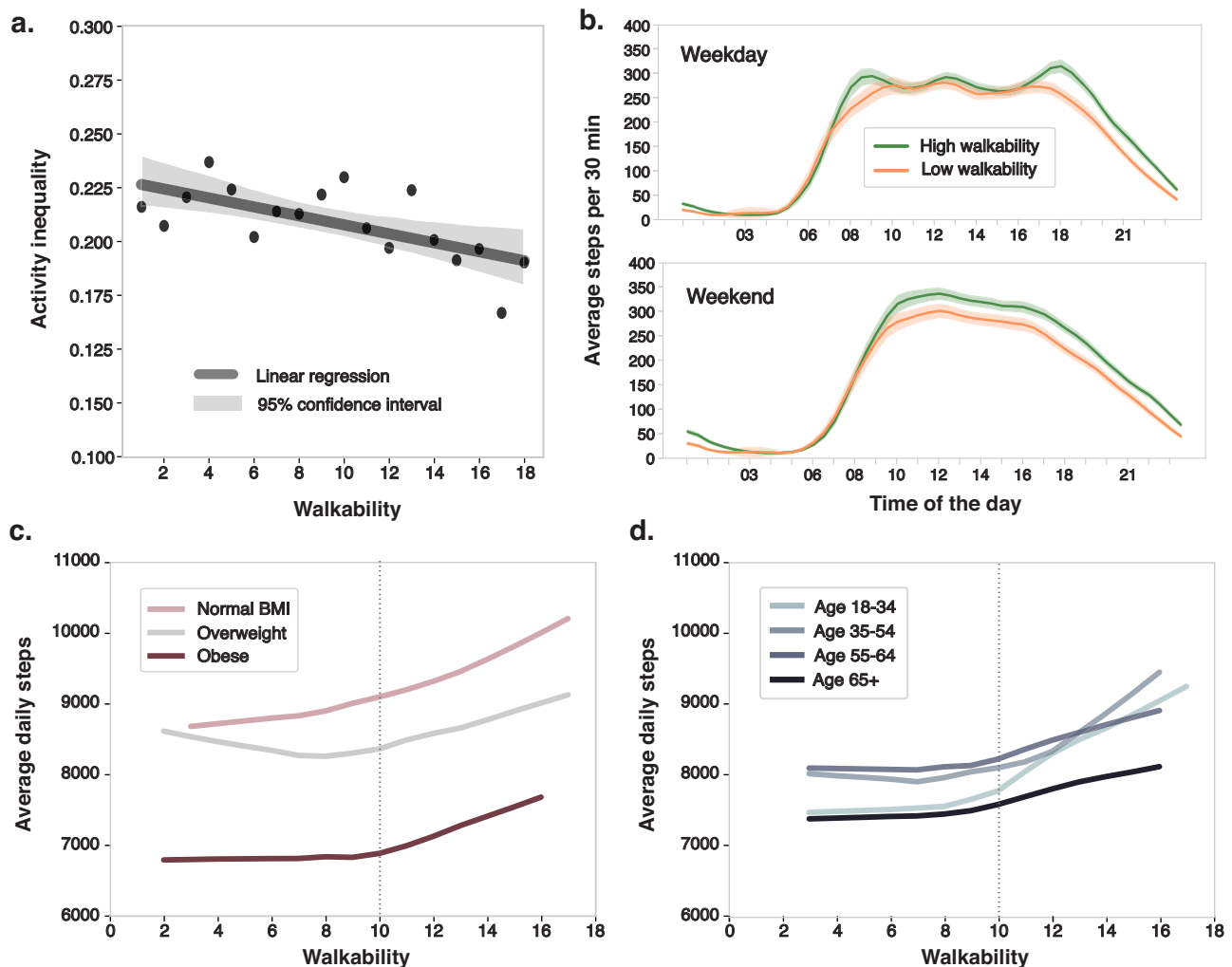


Fig. 4 | Relationship between perceived walkability and activity inequality and average daily steps. **a** (Figure adapted from Althoff et al. Figure 3⁵) Perceived Walkability and activity level measured by Fitbit. Analysis with the AoU dataset displays a linear relationship between activity inequality and walkability score as indicated in dark gray ($R^2 = 0.426$, $p < 0.01$). **b** (Figure adapted from Althoff et al. Figure 3⁵) Average steps were taken over the course of weekdays and weekends in the five most (indicated in green) and five least (orange) perceived walkability scores. **c** The walkability-activity gradient (WAG), which is measured by the slope of the linear regression model (y = average steps per day, x = perceived walkability) for

each subgroup, represents the predicted change in daily steps for a one-unit change in perceived walkability. The gray dotted line indicates the average perceived walkability score. Individuals with normal BMI consistently recorded the highest average steps, followed by those with overweight and obese BMI. WAG was also highest for normal BMI individuals when perceived walkability exceeded 10. **d** For all age groups, the highest WAG was observed beyond a perceived walkability score of 10. However, the increase in steps was less pronounced in older adults compared to younger groups.

Study population

The initial participant cohort was defined using the Cohort Builder tool on the AoU Researcher Workbench and consisted of 15,620 individuals who provided any Fitbit data (from Jan 2009 to Dec 2022) using their device. While AoU participant enrollment began in May 2018, historic Fitbit data before enrollment were made available. From this group, we selected 8886 participants who met the following criteria: had completed The Basics survey at the time of enrollment (survey allowed partial responses), at least one in-person physical measurement visit, provided any Fitbit activity and heart rate intraday data between January 2016 and December 2019, and age ≥ 18 (Supplementary Fig. 1). After excluding participants who do not meet the device adherence threshold (as defined in Feature Selection and Data Preprocessing) or participants who have exceedingly high average step counts (i.e., outliers defined by z -score > 4), we performed all analyses with 7403 participants. Of note, although available with the Controlled Tier dataset v7 (C2022Q4R11), we excluded any Fitbit data collected after the year 2020 to minimize the impact of the COVID-19 pandemic on activity behavior²⁷.

Feature selection and data processing

Demographic information, including date of birth, geolocation (state), and gender (self-reported as sex assigned at birth), was obtained from the responses to the basic survey at the time of enrollment. Body mass index (BMI) was extracted from each participant's initial in-person physical measurement visit and was used to screen participants for obesity according to the CDC guidelines²⁸ (18.5 to < 25 : normal, 25.0 to < 30 : overweight, ≥ 30 : obesity). Further, in this dataset, to protect participant privacy, birth dates were abstracted to year (i.e., set to the 15th of June of a participant's birth year)²⁹. Each participant's age was calculated relative to each of the data points available in the daily activity summary dataset, which was later averaged for the overall analysis. We also processed AoU's Social Determinants of Health (SDOH) survey that was launched in November 2021 to obtain information on participants' perceptions of their built environment. Specifically, we focused on the eight questions derived from the Physical Activity Neighborhood Environment Scale (PANES)³⁰ (Supplementary Table 1). When comparing our results by region, we followed the region designations provided by the U.S. Census Bureau³¹.

Fitbit data included daily summaries of activity (e.g., active minutes per day and total steps per day) and intraday data for heart rate and step counts. To ensure the validity of our statistical analysis, we implemented a heart rate filtering algorithm to identify days with suitable device adherence³². As Fitbit's intraday heart rate data is reported every minute, we aggregated all heart rate measurements from a given day per individual to determine the daily adherence rate, which is calculated by dividing the total number of heart rate measurements by the total number of minutes in a day (1440 min). We then plotted the daily adherence rate (ranging from 0 to 1) from all participants. From a visual inspection (Supplementary Fig. 2), we selected an adherence threshold (0.4) to label the valid days to be included in subsequent analyses. In selecting our optimal adherence threshold for the heart rate filtering algorithm, we mainly considered (1) minimizing data loss and conserving variance (i.e., the demographic distribution of participants) and (2) providing comparative results to Althoff et al.'s study⁵.

From further observation, we identified unreasonably high daily step values (i.e., daily steps $\geq 50,000$ steps, which is more than would be expected from a marathon run) from 172 unique individuals. We removed these days ($n = 1479$), attributing such deviations to errors in the device or its data processing. In addition, since the AoU database contains participants' historical Fitbit data from before their enrollment, any data points from before the participant turned 18 in age were removed from the analysis. Using all valid days, we calculated the average daily steps and active minutes for each participant and used z -scores to exclude any individuals with extreme outliers ($n < 20$), which are most likely due to having fewer valid days (defined as $|z\text{-score}| > 4$).

Lastly, an indicator of physical activity was derived based on the aerobic guideline by the U.S. Department of Health and Human Services. As defined by the Physical Activity Guidelines for Americans, an active individual engages in a minimum of 150 min of moderate-to-vigorous physical activity (MVPA) or 75 min of vigorous physical activity (VPA) per week¹. While the BRFSS¹⁰ determines whether individuals meet the aerobic activity guideline based on self-reported physical activity frequency and durations per week, here, we were able to leverage the "fairly active minutes" and "very active minutes" from Fitbit's daily summary report to derive weekly moderate-intensity and vigorous-intensity activity minutes, respectively. We consider an individual to have met the aerobic activity guideline if their fairly active minutes + 2*(vigorous-intensity activity minutes) ≥ 150 min/week¹.

Statistical analysis

To examine the relationship between activity inequality and various factors such as geographic location, biological sex, obesity, and walkability (an environmental factor), we used the methods described in Althoff et al.'s paper. This includes the determination of the Gini coefficient based on the Lorenz curve drawn by Althoff et al., which plots the proportion of the total daily steps of the population that are cumulatively recorded by the bottom X % of the population, where X is some pre-set threshold value⁵. The Gini coefficient is mathematically calculated by dividing the area that lies between the line of equality and the Lorenz curve by the total area under the line of equality³³. Thus, the Gini coefficient can be interpreted as an indicator of how dispersed the data is—values closer to 1 indicate higher inequality (or variance) within the population.

To investigate the geographic disparity in activity, we created a forest plot to visualize the mean of all individuals' daily steps in each state, along with a 90% confidence interval (Fig. 1a and Supplementary Table 2). Only states with >20 participants were included as per the data privacy and protection guidelines from the AoU Researcher Workbench. We also indicated the average annual temperature of each from January 2016 to December 2019, as obtained from the NOAA National Centers for Environmental Information³⁴. The activity trend was further compared to the mean results of the phone surveys conducted by BRFSS in 2017 and 2019 (Supplementary Fig. 3). We also generated a continuous probability density plot of daily steps for the two most active (New York and Illinois) and two least active states (Texas and Arizona) with more than 150 participants using the kernel density estimate (KDE) method³⁵ (Fig. 1b). The bandwidth

was selected using Scott's rule as the estimator and adjusted to smoothen the kernel density function. In addition, we normalized the density plot to each state's mode (representative of the baseline activity of each state) to better characterize the activity variance among individuals in each state (Fig. 1c).

The activity variance in a state was further investigated by plotting the activity of males and females separately to assess whether the activity differences across states (Fig. 1b) are related to sex assigned at birth (Fig. 2a). The association between activity inequality and gaps in activity by sex at birth, which is defined by the ratio of average daily steps of females to the difference in average daily steps between males and females, among states was also examined (Fig. 2b). We also examined if the gaps in activity by sex at birth differ depending on the typical physical activity level. We computed the first, second, and third quartiles of average steps per day in https://doi.org/10.1007/3-540-27661-0_4 each state (Fig. 2c).

For the analysis to understand the relationship between activity inequality and obesity, we only included states with >30 participants (Fig. 3a). A greater number of participants per state (compared to Fig. 1a) ensured sufficient data points to compute the activity inequality measure. We fit two linear regression lines that minimize the residual sum of squares—one with and one excluding outliers (Washington and Louisiana). We then investigated the association between sex assigned at birth and obesity by plotting the obesity prevalence (fraction of the population classified as obese) of each state against their average daily steps (Fig. 3b). The obesity prevalence and daily steps for each state were rounded to the nearest 1000.

Analysis of walkability and activity inequality was performed by defining "perceived walkability" from the relevant responses to AoU's SDOH survey (Supplementary Table 1). Out of the eight questions that were sourced from PANES³⁰, we eliminated two questions (Recreational Facilities and Day Safety) due to the low response rate (Supplementary Fig. 4). Apart from the question regarding the main housing type in the neighborhood, responses were transformed using a four-point scale, ranging from 0 (strongly disagree) to 3 (strongly agree), to gauge agreement with the statements related to walkability features. The questions regarding housing types were encoded on the same scale and by low (detached single-family housing) to high residential density (apartments or condos) (Supplementary Table 1). Perceived walkability was obtained by summing the scores from all six responses.

To determine the association of activity inequality with perceived walkability, we grouped individuals based on the perceived walkability of their neighborhood (Fig. 4a). To ensure statistical robustness, we excluded perceived walkability groups with ≤ 20 individuals ($n = 18$ groups included, $n = 1$ excluded). To explore activity trends in neighborhoods with high and low perceived walkability, we plotted Fitbit intraday step count over the course of the day. We binned the data into 30-min segments and averaged the data in each bin by weekdays or weekends. We plotted these 30-min-averaged steps for the five neighborhoods with the highest perceived walkability scores and the five neighborhoods with the lowest perceived walkability scores (Figs. 4b and Fig. 3c). We smoothed the curve by performing a moving average (taking the average of the last two 30-min bins) to better identify the diurnal trend. Finally, we examined the association between perceived walkability and activity level in each of the following subpopulations: sex assigned at birth, age, and obesity level (Fig. 4d). We again excluded groups with ≤ 20 participants. From the groups that met these criteria, we fit a LOESS (locally estimated scatterplot smoothing) regression model with perceived walkability score as the independent variable and daily steps as the dependent variable for each subpopulation. We compared the slopes, which represent the estimated increase in daily steps for a one-point increase in perceived walkability score, across subgroups. The parameters of the linear regression model for each subgroup are also reported (Supplementary Table 3).

Data availability

To ensure participant privacy, data used for this study is available to approved researchers following registration, completion of ethics training, and attestation of a data use agreement through the *All of Us* Research

Workbench platform, which can be accessed via <https://workbench.researchallofus.org/login>.

Code availability

All the registered Researcher Workbench users can access the code for this project at <https://workbench.researchallofus.org/library>.

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Author contributions

H.J., A.R., H.M., and J.K. had full access to the data in the study and took responsibility for the integrity of the data and the accuracy of the data analysis. Concept and design: H.J., A.R., J.K., J.D. Acquisition, analysis, or interpretation of data: H.J., J.K., A.R. Code review: A.R., H.M., A.K. Drafting of the manuscript: H.J., A.R., H.M., J.K., J.D. Critical revision of the manuscript for important intellectual content: H.J., A.R., H.M., J.D., J.K., A.K., P.H., M.B., K.M. Statistical analysis: H.J., A.R., J.K. Obtained funding: J.D., P.H., M.B. Administrative, technical, or material support: H.M., J.D. Supervision: J.D.

Competing interests

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Additional information

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